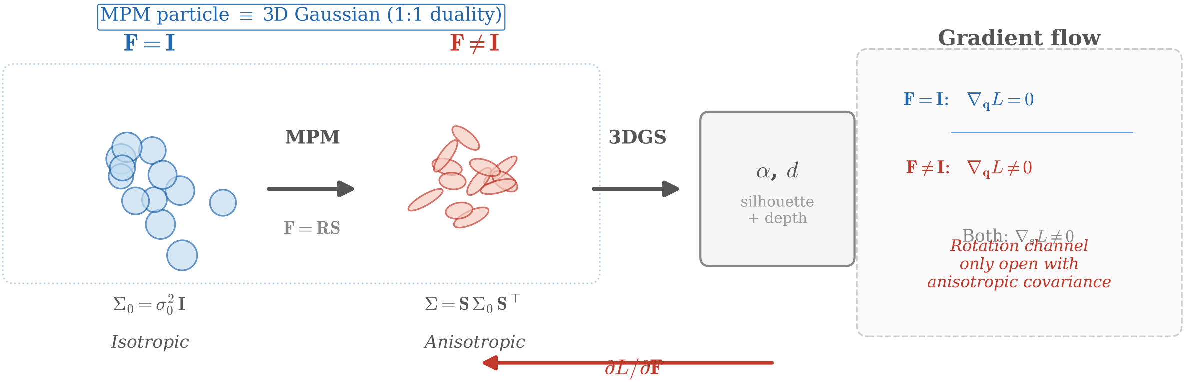

MPM particle ≡ 3D Gaussian (1:1 duality)
$\mathbf{F}=\mathbf{I}$
$\mathbf{F}\neq\mathbf{I}$
Gradient flow
MPM
$\mathbf{F}=\mathbf{RS}$
3DGS
α, d
silhouette
+ depth
$\mathbf{F}=\mathbf{I}$: $\nabla_{\mathbf{q}}L=0$
$\mathbf{F}\neq\mathbf{I}$: $\nabla_{\mathbf{q}}L\neq 0$
Both: $\nabla_{\mathbf{s}}L\neq 0$
Rotation channel
only open with
anisotropic covariance
$\Sigma_0=\sigma_0^2\,\mathbf{I}$
$\Sigma=\mathbf{S}\,\Sigma_0\,\mathbf{S}^\top$
Isotropic
Anisotropic
$\partial L/\partial\mathbf{F}$